3- Lab Tasks

3.1. DC measurements

1. Firstly we have constructed circuit 2.1. from the lab manual.

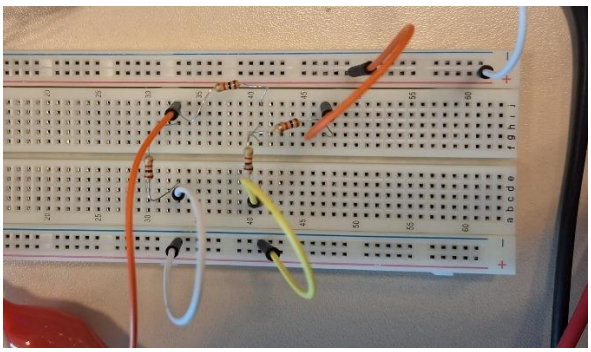


Figure 6 Circuit 2.1 on breadboard.

2. When we measure the same points on LTSpice and on breadboard, we get that table below the pictures of measurements.

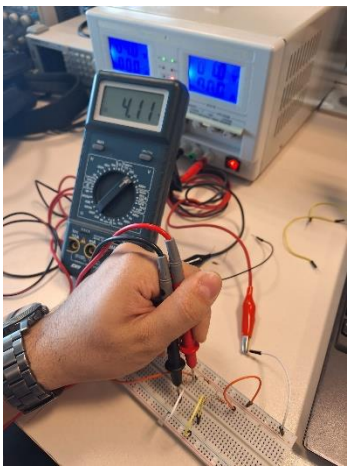


Figure 7 Measurements of node A

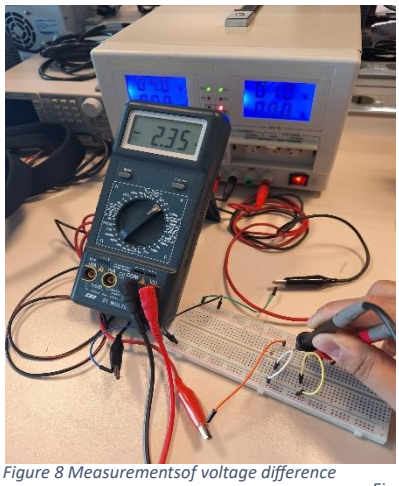


Figure 8 Measurementsof voltage difference on the R2.

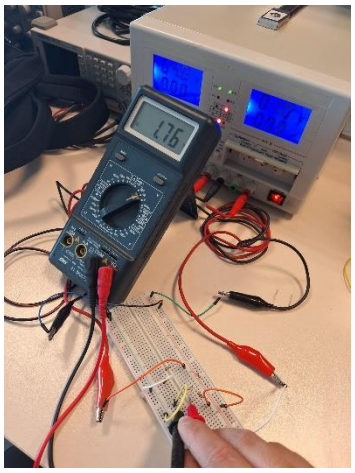


Figure 9 Measurements of node B

	On The LTSpice	In the lab measurements
A	4V	4.11V
B	1.67V	1.76V
I(R2)	-2.34mA	-2.35mA

We have measured the voltage differences between two side of the resistor and we calculated the current passes through the resistor.

I=V/R is our formula to find current. So we have got (-2.35V) on the multimeter that means I=(-2.35)/1000 and that gives -2.35 mAmp for the current passes through the resistor (R2).

There is some differences between LTSpice results and Lab measurements, but that is a negligible difference. They are happening because of some factors such as temperature, handshakes while measurement etc.

3.2 I-V curve

1. Firstly we have constructed circuit 3.1. from the lab manual. That circuit has 1V input and 10k resistor only.

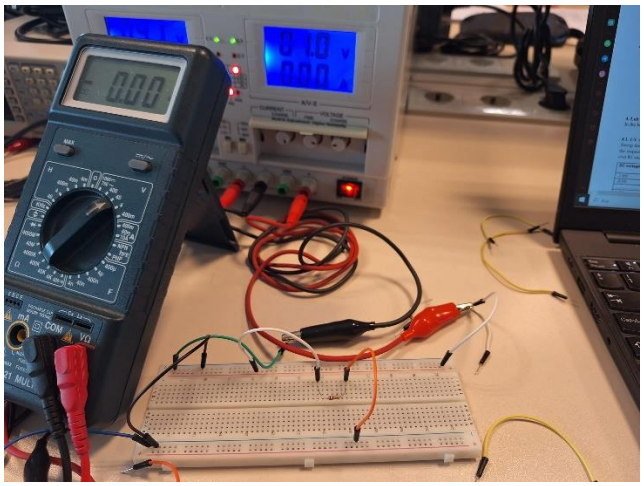
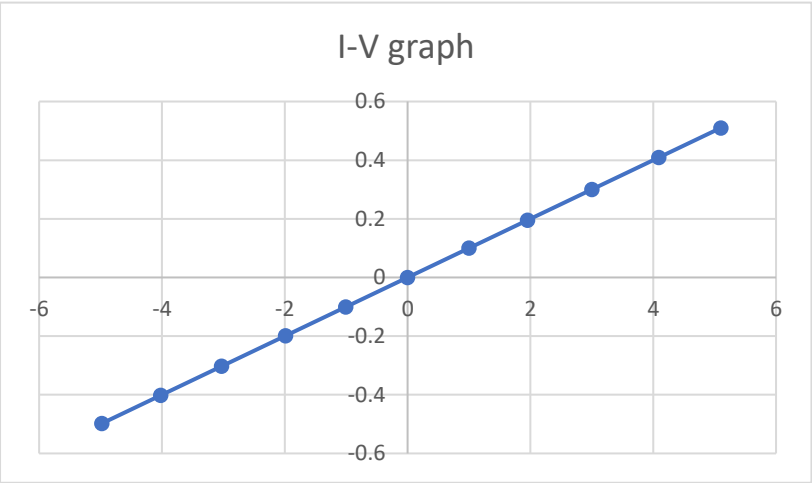


Figure 10 Circuit 3.1 on the breadboard.

3. In that task we analyze the electrical properties of a 10k ohm resistor when subjected to varying voltage levels within the range of 5V to -5V, incrementally adjusted in 1V intervals. Then we measure Voltage differences on each side of 10k resistor.

   All of our measurements for the circuit 3 is on the table, shown below.

V	Measured V with multimeter (V)	I=V/R (mAmp) (R=10k)
-5	-4.98	-0.498
-4	-4.02	-0.402
-3	-3.03	-0.303
-2	-1.99	-0.199
-1	-1.01	-0.101
0	0.00	0.00
1	1.00	0.100
2	1.95	0.195
3	3.00	0.300
4	4.09	0.409
5	5.10	0.510



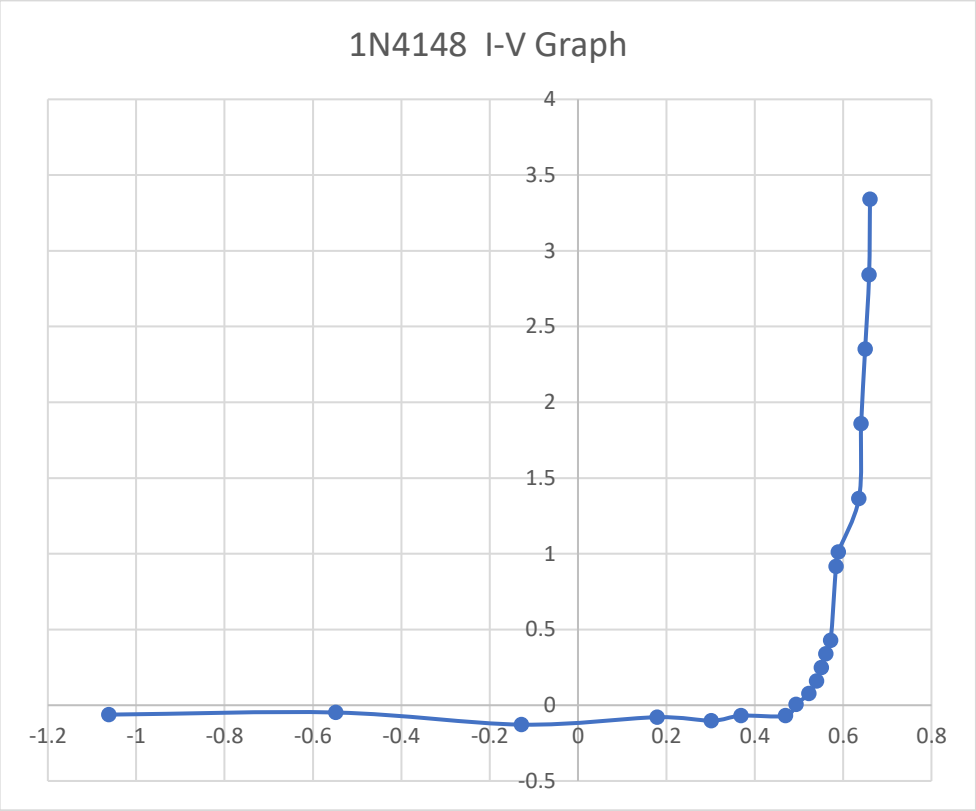
4. We can observe that Current and Voltage values are directly proportional to each other. Also, there is not any curve for the resistor on the I-V graph.

4- Lab Tasks

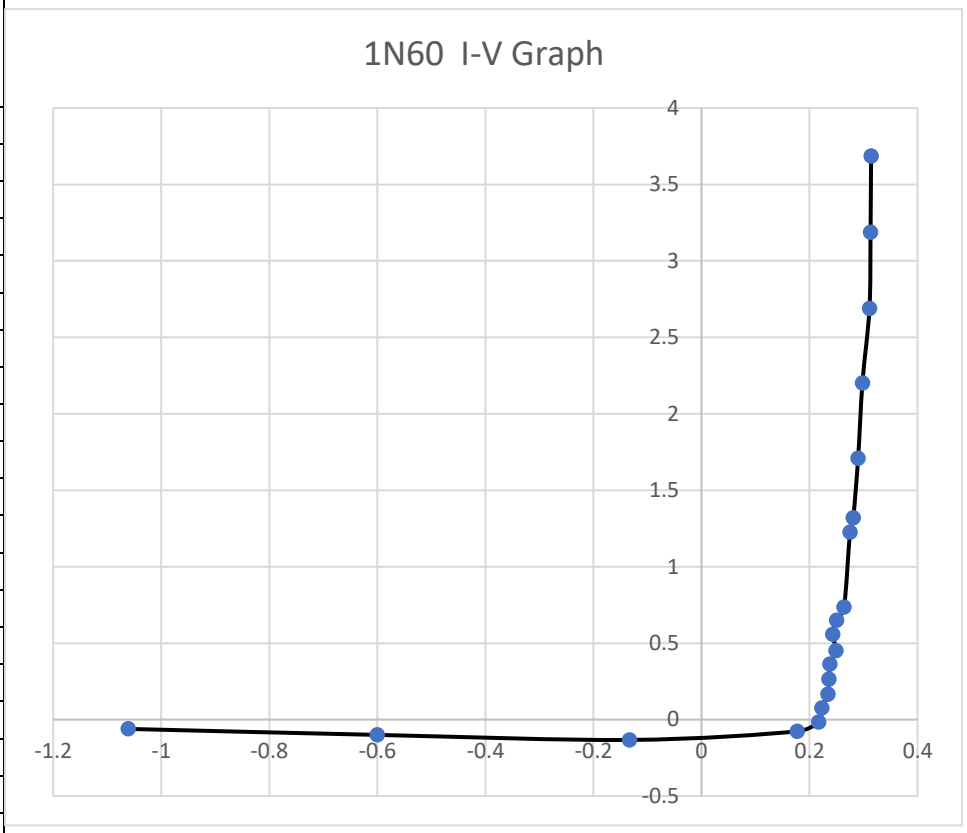
4.1. I-V curves

1. For the task 4, we will look at the how different diodes behave differently. We have tried diodes 1N4148, 1N60, Zener diodes and LED to observe their behaviors as a diode. To analyze the behaviors we have built the circuit 2.1 as shown in the lab manual.
2. Next, we measured the Voltage difference on the Resistor with the multimeter, then we have changed voltage between -1V to 4V (Except Zener, we have used -10V to 0V because Zener diode creates extra reverse breakdown voltage.)

1N4148			
DC Voltage (V)	R1 Voltage	D1 Voltage	D1 Current (mA)
-1	0.00	-1.06	-0.06
-0.5	0.00	-0.55	-0.05
0	0.00	-0.13	-0.13
0.1	0.00	0.18	-0.08
0.2	0.00	0.30	-0.10
0.3	0.00	0.37	-0.07
0.4	0.05	0.47	-0.07
0.5	0.10	0.49	0.01
0.6	0.17	0.52	0.08
0.7	0.28	0.55	0.15
0.8	0.33	0.55	0.25
0.9	0.43	0.56	0.34
1	0.51	0.57	0.43
1.5	1.03	0.58	0.92
1.6	1.04	0.59	1.01
2	1.45	0.64	1.36
2.5	1.98	0.64	1.86
3	2.40	0.65	2.35
3.5	2.48	0.66	2.84
4	3.28	0.66	3.34



1N60			
DC Voltage (V)	R1 Voltage	D1 Voltage	D1 Current (mA)
-1	0.00	-1.06	-0.06
-0.5	0.00	-0.60	-0.10
0	0.00	-0.13	-0.13
0.1	0.00	0.18	-0.08
0.2	0.09	0.22	-0.02
0.3	0.11	0.23	0.08
0.4	0.15	0.23	0.17
0.5	0.20	0.23	0.27
0.6	0.30	0.24	0.37
0.7	0.33	0.25	0.45
0.8	0.57	0.24	0.56
0.9	0.65	0.25	0.65
1	0.76	0.26	0.74
1.5	1.22	0.28	1.22
1.6	1.36	0.28	1.32
2	1.68	0.29	1.71
2.5	2.17	0.30	2.21
3	2.68	0.31	2.69
3.5	2.92	0.32	3.18
4	3.55	0.31	3.69



ZENER			
DC Voltage (V)	R1 Voltage	D1 Voltage	D1 Current (mA)
-10	-8.00	-0.74	-9.26
-9.5	-8.01	-0.74	-8.76
-9	-6.67	-0.74	-8.26
-8.5	-7.01	-0.75	-7.75
-8	-6.69	-0.75	-7.26
-7.5	-6.56	-0.75	-6.75
-7	-6.05	-0.74	-6.26
-6.5	-5.54	-0.74	-5.76
-6	-4.89	-0.73	-5.27
-5.5	-3.73	-0.73	-4.77
-5	-3.93	-0.73	-4.27
-4.5	-3.33	-0.72	-3.78
-4	-2.99	-0.72	-3.28
-3.5	-2.79	-0.72	-2.78
-3	-2.08	-0.72	-2.28
-2.5	-1.74	-0.72	-1.78
-2	-0.74	-0.66	-1.34
-1.5	-0.83	-0.68	-0.82
-1	-0.33	-0.61	-0.39

